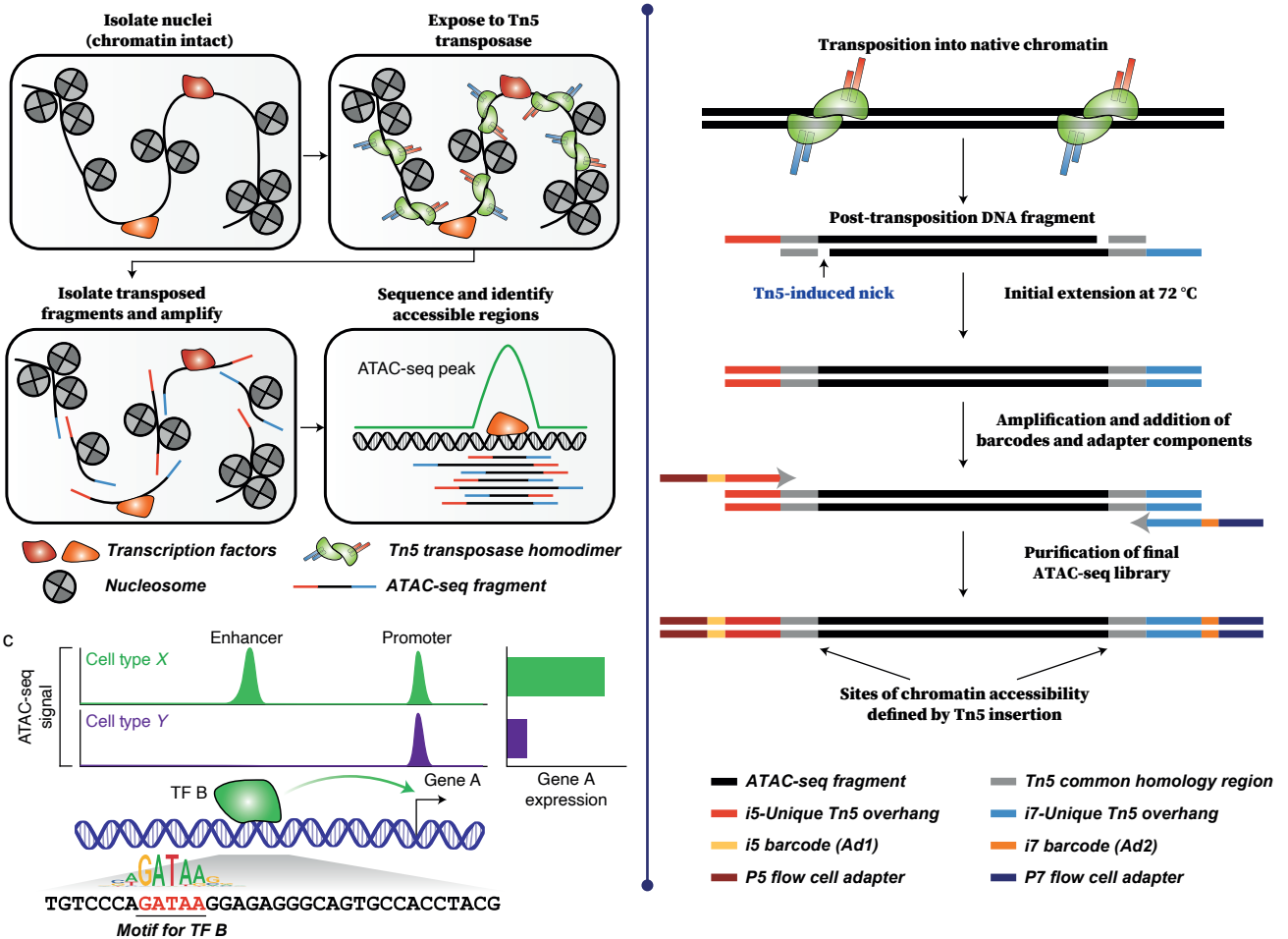

Isolate nuclei
(chromatin intact)
Expose to Tn5
transposase
Isolate transposed
fragments and amplify
Sequence and identify
accessible regions
ATAC-seq peak
Transcription factors
Nucleosome
Tn5 transposase homodimer
ATAC-seq fragment
c
Enhancer
Promoter
ATAC-seq
signal
Cell type X
Cell type Y
TF B
Gene A
Gene A
expression
TGTCCCAGATAAGGAGAGGGCAGTGCCACCTACG
Motif for TF B
Transposition into native chromatin
Post-transposition DNA fragment
Tn5-induced nick
Initial extension at 72 °C
Amplification and addition of
barcodes and adapter components
Purification of final
ATAC-seq library
Sites of chromatin accessibility
defined by Tn5 insertion
ATAC-seq fragment
Tn5 common homology region
i5-Unique Tn5 overhang
i7-Unique Tn5 overhang
i5 barcode (Ad1)
i7 barcode (Ad2)
P5 flow cell adapter
P7 flow cell adapter